Linux Log Analyser & Monitoring

In this project you will create a **log analyser** for Linux log files. Your program must read a real Linux log file, extract useful information, and generate a **monitoring summary**.
The key objective is that you decide what information is important and how it should be monitored. This project simulates real-world work in cloud infrastructure, system administration, and cybersecurity, where logs are used to detect problems and incidents.

1. Input Data

You must work with at least one Linux log file, for example:

- /var/log/syslog
- /var/log/auth.log
- /var/log/kern.log

or another log file approved by your teacher.

2. Parsing

You must extract information from log lines using **regular expressions (regex)**. Your parser should extract at least 3 fields from each log entry, for example:

- timestamp
- hostname
- program / service name (sshd, sudo, systemd, …)
- PID (if available)
- message text
- IP address or username (if present)

3. Monitoring & Analysis

You must design your own monitoring system. This means you must decide what is important and what should be measured. Your monitoring must include:

1. At least 2 statistics/counters, for example:
    - Top 10 programs/services producing log messages
    - Most common error messages or keywords
    - Top IP addresses (e.g. suspicious activity)
    - Number of failed login attempts
    - Number of warnings/errors per service

2. At least 1 time-based metric, for example:
    - events per minute
    - failed logins per 5 minutes

4. Alert Rule

Your program must implement at least one alert rule. An alert rule is a condition that triggers a warning, for example:

- more than 20 failed logins in 5 minutes
- more than 10 error messages in 1 minute
- one IP address triggers too many suspicious events
- too many “permission denied” messages in a short time

When the alert triggers, your program must print a clear output like:

```
ALERT: Suspicious activity detected (…reason…)
```

5. Output Requirements

Your program must output a clear monitoring summary in the terminal, including:

- total parsed lines
- total unparsed lines
- your chosen statistics
- your time-based metric results
- triggered alerts (if any)

6. Deliverables

You must submit two files. One is your Python source code and the other is documentation. The documentation should contain all essential information about which elements from the log file are examined and how your application works.